

AMSC5738 Surface Science and Thin Film Technology

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Class time: Mondays 6:30 pm–9:15 pm

Classroom: LSK LT2

We will use Blackboard. There will be no dedicated course website.

Assessment scheme

Tests (2×)	60%
Homework assignments (5×)	15%
Presentation (1×)	25%

Presentation

- One presentation
- Select topics from recent research papers (**provided**) that are related to thin films

Grading based on:

- ✓ Scientific understanding (50%)
- ✓ Slide preparation (25%)
- ✓ Oral presentation (25%)

References

- 1) Donald L. Smith, “Thin-Film Deposition: Principles & Practice”, McGraw-Hill, Inc., TA418.9.T45S65 1995.
- 2) Milton Ohring, “The Materials Science of Thin Films”, Academic Press, TA418.9.T45 O47 1992.
- 3) K. Oura, V. G. Lifshits, A. A. Saranin, A. V. Zotov, M. Katayama, “Surface Science – An Introduction”, Springer, QC173.4.S94 S96425 2003.
- 4) Anthony C. Jones, Michael L. Hitchman, Ed., “Chemical Vapor Deposition: Precursors, Processes and Applications”, RSC, TS695.C67 2009.
- 5) Hans Lüth, “Solid Surfaces, Interfaces and Thin Films”, 6th Edition, Springer, QC173.4 S94 L86 2015.
- 6) Terry L. Alford, Leonard C. Feldman, James W. Mayer, “Fundamentals of Nanoscale Film Analysis”, Springer, QC176.83 A44 2007.

Use of AI Tools

Approach 2 - Use only with prior permission

The use of AI tools is permitted when students study the course materials, work on the homework assignments, and prepare for their oral presentations. It is not allowed in the mid-term exam or final exam.

Use AI tools to learn knowledge and to improve understanding. Please DO NOT rely solely on AI to give you answers.

Some characteristics of this course

An experimental course

Contain many sub-topics, which are **not closely related**

Lack coherency among the contents

Emphasize how to apply learned knowledge to problem solving

Many qualitative descriptions and estimations, without rigorous derivations

Mainly based on lectures and notes

One experimental demonstration, but without assessment

Notes

Duration (7 Sept. 2026 – 5 Dec. 2026, 13 weeks, 12 classes)

Holiday (19 Sept. 2026, Chung Yeung Festival)

Demonstration of evaporation (end of Sept. 2026)

Mid-term (end of Oct. 2026; final: 7–10 Dec. 2026)

Oral presentations (1–4 Dec. 2026)

Many words on slides for better understanding

Notes (Continued)

An experimental course. Useful for those who work on thin films and solid materials, and those who are interested in knowing more about thin films and solid materials.

I highly suggest you work and submit homework assignments and give oral presentations. Their points are easy to get. The points for the two exams are not easy.

The exams will consist of multiple-choice questions and long questions. The former are taken from the lecture slides, and the latter are modified from questions in the homework assignments.